Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

Draft outline

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1 Introduction

Layered van der Waals solids and related two-dimensional materials are commonly characterized by wide-angle x-ray scattering (WAXS) with area detectors, because a single exposure maps a large reciprocal-space volume and directly reports both structure and orientation.[1–5] Orientation is often the primary performance variable in these anisotropic films.[6–8] In layered halides such as PbI_2, transport and device response can depend strongly on whether the layers lie parallel to the substrate.[9] In topological bismuth chalcogenides, strong c-axis texture is needed to preserve the desired electronic response.[10] In layered halides such as PbI_2, transport and device response can depend strongly on whether the layers lie parallel to the substrate.[9] In topological bismuth chalcogenides, strong c-axis texture is needed to preserve the desired electronic response.[10] In transition-metal dichalcogenides, catalytic activity increases when orientation exposes platelet edges.[11] Comparable 2D oriented-powder textures also occur across a wider set of layered thin-film families, including telluride chalcogenides (e.g., Bi_2Te_3, Sb_2Te_3), dichalcogenides (e.g., MoS_2, NbS_2), layered halides and oxyhalides (e.g., BiI_3, BiOI), and laminated 2D-flake films (e.g., $Ti_3C_2T_x$ MXenes and graphene-oxide papers).[12–19]

A common thin-film preferred orientation in these systems is a two-dimensional (2D) oriented powder where plate-like crystallites lie flat with aligned layer normals but remain azimuthally randomized in-plane.[8, 20, 21] This orientational state is neither a single crystal nor a fully isotropic three-dimensional (3D) powder. As a result, analysis strategies that assume discrete reciprocal-lattice points (single crystals) or spherical powder averaging (3D powders) can fail or become ambiguous for layered films in the intermediate regime.[20]

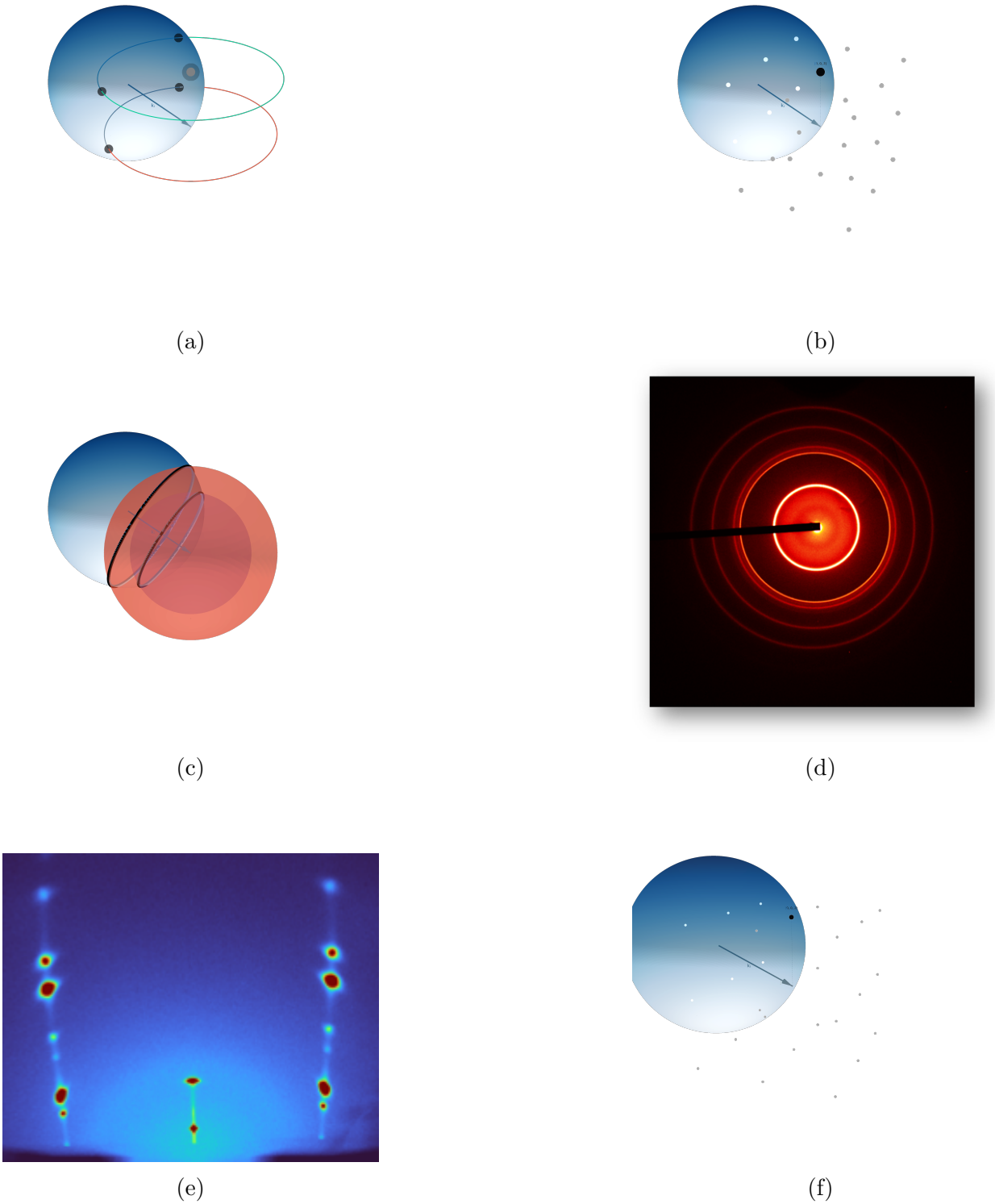

Figure 1: Diffraction limits for single crystals and powders. (a) 2D powder: cylindrical shells about the film normal (including specular $(00L)$); (b) single crystal: discrete points; (c) 3D powder: spherical shells. Detector signatures: (d) Debye–Scherrer rings; (e) specular caps and off-specular arcs; (f) discrete Bragg spots.

Despite this hybrid information content, many WAXS workflows reduce the area-detector image to 1D profiles by azimuthal integration, often with masking near the direct beam and specular region.[3, 5, 22–24] These steps are well suited to nearly isotropic powders, but for 2D powders they deweight or remove the arcs and specular caps that are most sensitive to out-of-plane correlations, stacking, and mosaic tails.[1, 3] Stacking disorder is also commonly inferred from 1D interference models and refinements,[25–27] which do not directly exploit the full two-dimensional anisotropy of oriented-film scattering. More general intermediate orientation distributions can be treated with ODF-based refinements or preferred-orientation corrections.[28–33] In WAXS, recovering a flexible ODF typically requires extensive reorientation data and added degrees of freedom.[34, 35] We therefore assume the physically motivated 2D-powder orientation distribution a priori (axial symmetry with a low-dimensional mosaicity kernel) and refine its parameters by forward-modeling the full 2D detector image, so that the arc and specular (Bragg-rod) line shapes in Fig. 1 act as direct constraints to reduce computational resources while improving fit precision.[1, 3]

Figure 1 summarizes the associated reciprocal-space and detector-space limits. In the Ewald construction (Fig. 1(a–c)), elastic scattering requires that reciprocal-space intensity lie on the Ewald sphere of radius $|\mathbf{k}|$.[36] For a single crystal, reciprocal space consists of isolated points (Fig. 1b), so at fixed incidence angle the Ewald sphere typically misses most reflections and the detector records only sparse spots for special orientations (Fig. 1f).[36] For a 3D powder, random orientations smear each Bragg point into a spherical shell (a Bragg sphere), and the Ewald sphere intersection produces Debye–Scherrer rings on the detector (Fig. 1c, Fig. 1d).[36] For a 2D powder, azimuthal disorder about the common c axis smears each off-specular Bragg point into an in-plane ring about Q_z, while specular $(00L)$ reflections remain localized near the Q_z axis (Fig. 1a).[1, 20] When the Ewald sphere intersects these features, the detector records hybrid signatures: pairs of symmetric spots from the ring intersections and single-crystal-like spots near the specular stripe from $(00L)$ intensity (Fig. 1e).[1, 3, 37] Finite mosaicity then redistributes intensity in polar angle, broadening the ring intersections into arcs and spreading the $(00L)$ spots into cap-like features near the specular region.[1, 20]

Here we present a quantitative forward model for WAXS from two-dimensional powders that predicts full area-detector images and refines them pixel by pixel against measured data.[34, 35, 38, 39] The model embeds calibrated detector geometry and grazing-incidence wave propagation within the forward calculation, and it parameterizes mosaicity and disorder explicitly so that reflection profiles in Fig. 1 act as direct constraints.[1, 5, 34, 35] We validate the ordered 2D powder limit using MBE-grown Bi_2Se_3 and Bi_2Te_3 films, and we then apply the same framework to disordered PbI_2 to extract stacking-fault parameters from crystal truncation rods along G_z that are typically masked in 1D-integrated analyses (see supporting information, Fig. S1 and Secs. S2–S3).[9, 10, 26, 27] By embedding geometric and sample effects within a single self-consistent forward model, we reduce correction-induced bias and enable joint refinement of geometry, mosaicity, and disorder directly from the raw images.[3, 24, 39]

2 Diffraction Model

We model the measured area-detector intensity by combining a calibrated instrument geometry with a crystallographic scattering model and a low-dimensional mosaicity spread appropriate to two-dimensional powders. The forward calculation enumerates allowed reciprocal-lattice vectors $\mathbf{G}$, samples rays from the finite-divergence incident beam, applies the diffraction condition, and propagates diffracted rays to the detector so that simulated images can be compared to experiment on a pixel-by-pixel basis. The essential elements are introduced below: (i) the beam, sample, and detector geometry that defines the pixel-to-$\mathbf{Q}$ mapping; (ii) the crystallographic structure factors that set reflection intensities; and (iii) the mosaicity model that redistributes intensity about each ideal Bragg condition.

2.1 Diffraction Geometry

In the lab frame, the incident wavevector $\mathbf{k}_i$ propagates along the y axis, with a product of two Gaussian profiles in the x and z directions, as shown in Fig. 2. We treat the incident beam as a distribution in transverse position (widths w_x, w_z) and direction (rms divergences $\Delta\theta_x, \Delta\theta_z$). A Monte Carlo ray-tracing approach samples rays from this beam phase space and propagates each ray to the sample and then to the detector; simulated detector images are formed by accumulation over these rays.

The goniometer defines a rotation axis about which the sample normal rotates, as shown in Fig. 3. In the aligned configuration, a rectangular film of in-plane dimensions W and H lies in the x-y plane with thickness t along the sample normal. The rotation axis lies within this plane, with the incident beam parallel to the x-y plane and perpendicular to the rotation axis. In practice, the beam center and sample center may be displaced from the goniometer axis by vertical offsets z_B and z_S, respectively. Rotating the sample about this axis sets the incidence angle θ_i between the incident wavevector $\mathbf{k}_i$ and the sample surface, as shown in Fig. 2. If the substrate is not perfectly parallel to the x-y plane (e.g., it is not flush against the sample holder), the sample normal can also be tilted by a residual angle δ. For each trial ray, we compute its intersection with the sample plane. The finite beam size at grazing incidence produces an elongated footprint of length X_s (Fig. 3); rays that fall outside the finite illuminated window set by W and H miss the sample and are discarded. Because rays are sampled from the spatial beam profile, partial illumination and footprint truncation are naturally captured.

For rays that intersect the film, wave attenuation occurs both at the surface (Fresnel transmission) and during propagation through the medium (absorption). We model the film by a complex refractive index $\tilde{n} = 1 - \delta_n + i\beta_n$, compute transmitted internal wavevectors using Snell's law together with continuity of the in-plane components, and use the resulting complex out-of-plane components to set the attenuation along the incident and exit paths through thickness t. Full expressions for the internal wavevectors, Fresnel factors, and thickness-averaged absorption weight are given in the supporting information (Sec. S1.2).

The 2D area detector is positioned at a distance D from the sample along the lab y axis, with tilts about the x and z axes denoted β and γ, respectively, as shown in Fig. 2. An additional in-plane detector rotation χ accounts for rotation of the detector axes within the detector plane. A diffracted ray with exit wavevector $\mathbf{k}_f$ in the lab frame intersects the detector at a scattering angle 2θ, defined as the angle between $\mathbf{k}_f$ and the y axis, and at a detector-plane azimuth ϕ, measured from the detector x' axis.

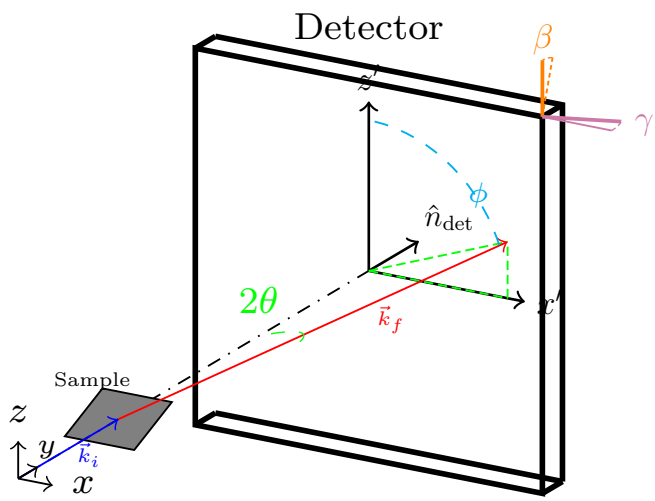

Figure 2: Overall lab-frame geometry. The incident beam $\mathbf{k}_i$ propagates along y with widths w_x, w_z and divergences $\Delta\theta_x, \Delta\theta_z$. The detector at distance D is tilted by β and γ, and referenced to offsets (x_0, y_0). A representative $\mathbf{k}_f$ defines the scattering angle 2θ and detector-plane azimuth ϕ.

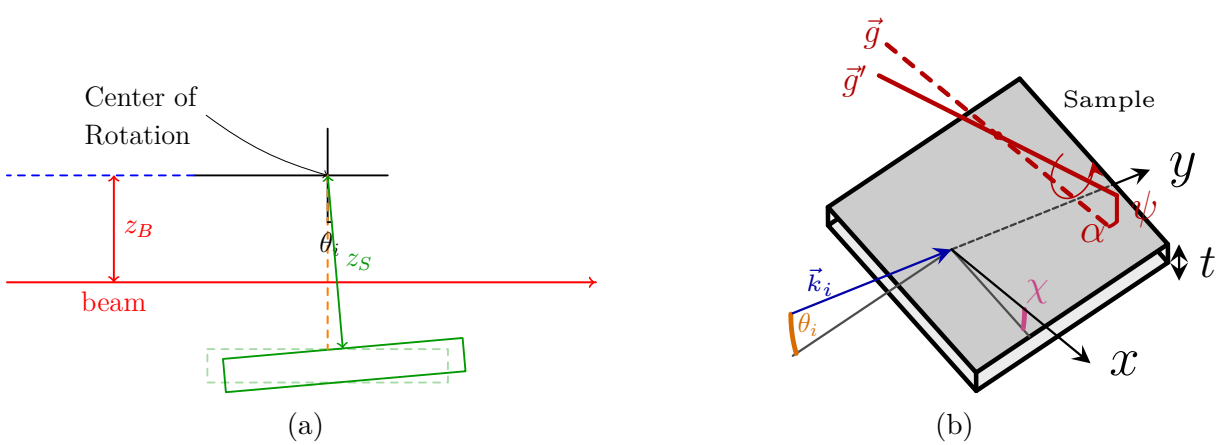

Figure 3: Two-stage sample geometry. Left: rotation about the goniometer center with sample offset z_S and beam offset z_B. θ_i is the incidence angle. Right: full alignment adds tilt δ and footprint length X_s within an illuminated window (W, H) on a film of thickness t. $\vec{g}$ is the goniometer arm (normal to $\hat{y}$), and $\vec{g}'$ is the misaligned arm obtained by rotations α and ψ.

2.2 Structure Factor

Quantitative refinement of two-dimensional X-ray diffraction images requires a crystal structure model, which determines the structure factors $F_{\mathbf{G}}$ and the associated set of reciprocal-lattice vectors $\mathbf{G}$. For a given reflection, the magnitude $|F_{\mathbf{G}}|$ governs the scattered intensity, while $\mathbf{G}$ sets the diffraction condition through the intersection of the Ewald sphere with reciprocal space. We therefore enumerate all $\mathbf{G}$ whose scattering angle 2θ falls within the detector acceptance. Standard approaches reduce cost by grouping symmetry-equivalent reflections via a multiplicity factor and scaling the intensity accordingly. For two-dimensional powders, where the experimental geometry and orientation distribution can make multiplicity nontrivial and structure-dependent, we instead evaluate every allowed $\mathbf{G}$ explicitly. For each $\mathbf{G}$ we compute $|F_{\mathbf{G}}|^2$ together with the corresponding exit wavevector $\mathbf{k}_f$, so that multiplicity and orientation effects emerge naturally within the forward model.

For each reflection with reciprocal-lattice vector

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*,$$

we compute the structure factor as

$$F_{\mathbf{G}} = \sum_a o_a \, f_a(Q) \, \exp[2\pi i \, \mathbf{G} \cdot \mathbf{r}_a] \, \exp\Big[-2\pi^2 \, \mathbf{G} \cdot \mathbf{U}_a \cdot \mathbf{G}\Big] \,, \tag{1}$$

where the sum runs over atoms a in the unit cell, o_a is the site occupancy, $f_a(Q)$ is the atomic form factor with $Q = |\mathbf{G}|$, and $\mathbf{r}_a$ is the fractional atomic position set by the space-group symmetry (Wyckoff site). If we refine positions, we vary only the free parameters of the occupied Wyckoff sites, not independent per-atom shifts. The Debye–Waller factor is written in anisotropic tensor form, where $\mathbf{U}_a$ is the mean-square displacement tensor (ADP) for atom a. In refinement, $\mathbf{U}_a$ may be constrained by symmetry or reduced parameterizations when warranted by the data.

2.3 Mosaicity

Imperfect 2D powders exhibit out-of-plane mosaicity, with crystallite layer-normals (van der Waals crystal normals) distributed about the substrate normal (Fig. 4).

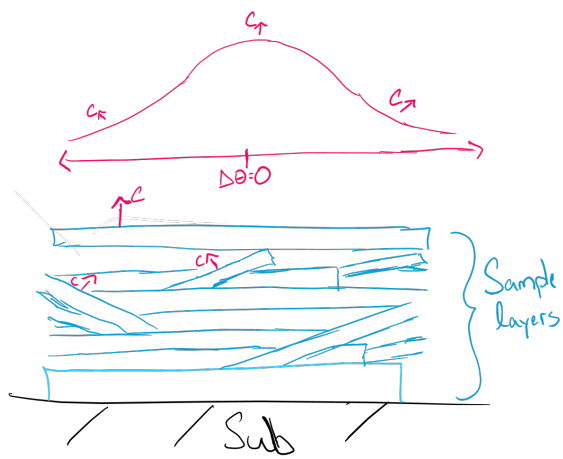

Figure 4: Schematic of crystallite layer-normal misorientation (mosaicity). A distribution of crystallite orientations may form in a thin film.

We describe out-of-plane mosaicity by a misorientation density $\omega(\Delta\theta)$, where $\Delta\theta$ is the crystallite tilt angle relative to the substrate normal ($\Delta\theta = 0$ for perfect alignment). Because orientations are periodic, we treat ω as a 2π-periodic wrapped distribution (see supporting information, Sec. S1.1). We model ω as a unit-normalized sum of a Lorentzian and a Gaussian:

$$\omega(\Delta\theta) = \eta \frac{1}{\pi} \frac{\Gamma}{\Delta\theta^2 + \Gamma^2} + (1-\eta) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\Delta\theta^2}{2\sigma^2}\right), \qquad 0 \le \eta \le 1, \tag{2}$$

with Lorentzian half-width at half maximum Γ and Gaussian standard deviation σ. The inclusion of the Lorentzian component is not immediately intuitive and will be explained in more detail later in the discussion.

For a reflection, we work in the meridional section (G_r, G_z) of reciprocal space, where G_r is the projection of the reciprocal lattice vector onto the plane of the substrate, and G_z is the component along the surface normal. These components are related to the full reciprocal lattice vector $\boldsymbol{G} = (H, K, L)$, but the relationship depends on the crystal symmetry. For a general crystal, G_r is defined as the magnitude of the projection of $\boldsymbol{G}$ onto the plane, and G_z is the component along the normal direction.

The analysis is motivated by considering rotations about the substrate surface normal, allowing for a description of mosaicity without tying directly to H, K, L coordinates, but rather focusing on the resulting distribution in reciprocal space with line density $I_0\omega(\Delta\theta)$, where I_0 is the total intensity associated with a given reflection prior to redistribution by mosaicity.

Figure 5 summarizes how this reciprocal-space broadening appears in the meridional $\{G_r, G_z\}$ plane and how it maps onto characteristic detector-space features for in-plane and specular conditions.

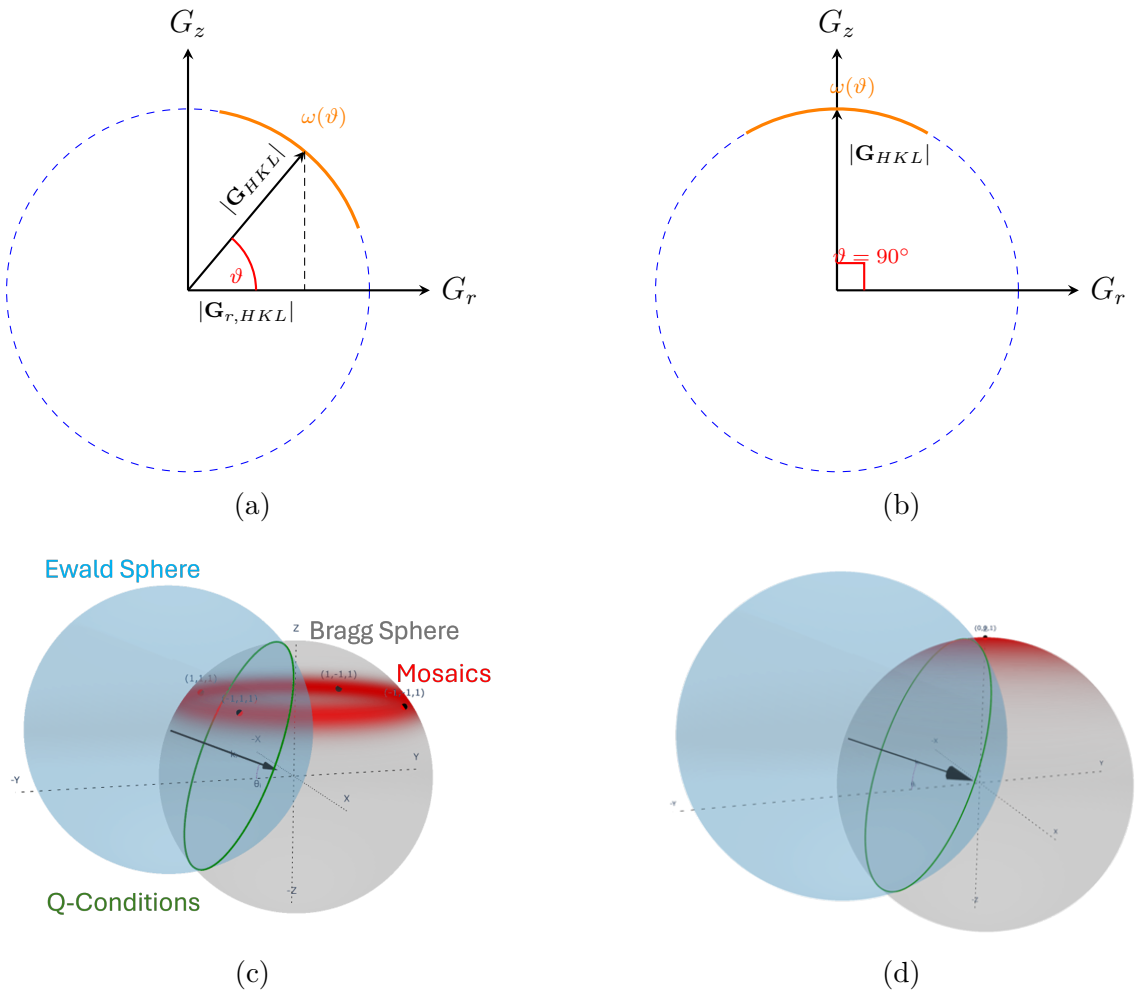

Figure 5: Mosaic broadening viewed in the $\{G_r, G_z\}$ plane (a,b) and its detector-space manifestation (c,d). Intensity is redistributed along the Bragg circle with line density $I_0\,\omega(\Delta\theta)$. The in-plane condition with $G_r \neq 0$ (a,c) yields two arcs of reflection, while the specular reflection with $G_r = 0$ (b,d) concentrates intensity near the polar cap, forming a single arc.

Random in-plane rotations about the film normal revolve the meridional line density about the G_z axis, generating an axially symmetric distribution on the Bragg sphere.

We normalize the corresponding Bragg-sphere surface density $\sigma(\vartheta)$ so that $\int \sigma(\vartheta)\, dA = I_0$, yielding

$$\sigma(\vartheta) = \frac{I_0}{2\pi R^2 \sin\vartheta}\,\omega(\vartheta), \qquad 0 < \vartheta < \pi, \tag{3}$$

where R is the Bragg-sphere radius for the reflection. The factor $1/\sin\vartheta$ is the Jacobian for converting the one-dimensional tilt distribution ω into a surface density.

The reciprocal-space consequences appear on an area detector in Fig. 6. Strong out-of-plane mosaicity concentrates specular intensity near discrete G_z values at $G_r = 0$ (region A), while random in-plane orientation distributes in-plane reflections uniformly in azimuth about G_z, producing ring-like features (region B).

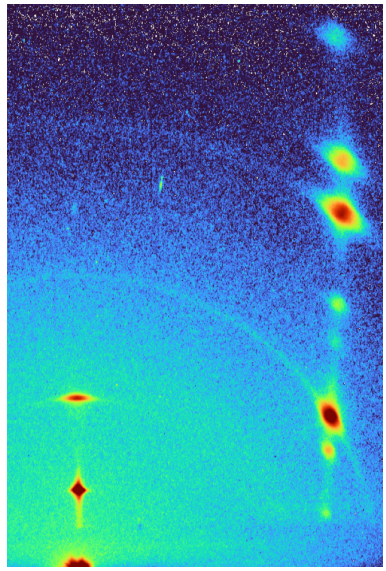

Figure 6: Log-scale WAXS pattern of Bi_2Se_3 at $\theta_i = 5°$. Region A: specular reflections. Region B: in-plane scattering.

In the in-plane region (Fig. 6B), the Ewald sphere intersects the broadened Bragg-sphere feature along an extended band. Azimuthal averaging then produces band-like scattering whose radial width and line shape are set primarily by the core of ω and by the Jacobian factor $1/\sin\vartheta$ in Eq. (3). In the specular region (Fig. 6A), the same mosaic spread produces cap-like broadening about the specular reflections. Off-condition specular intensity arises from crystallites whose layer normals are tilted by $\Delta\theta$ away from the mean, so the decay of intensity away from the nominal condition is governed primarily by the tails of ω (with only weak variation from the geometric factor in Eq. (3)).

This observation explains why multiple specular reflections can remain visible for a single incident $\mathbf{k}_i$, even when their nominal θ_i conditions lie outside the measurement window. If ω contains a Lorentzian component ($\eta > 0$), then for $|\Delta\theta| \gg \Gamma$,

$$\omega(\Delta\theta) \sim \eta \, \frac{\Gamma}{\pi \, \Delta\theta^2},$$

so the intensity decays algebraically rather than exponentially. In practice this makes the specular reflections a strong constraint during refinement, since peak persistence and line shape at large $\Delta\theta$ directly determine both the width and the tail character of ω, while in-plane rings are less sensitive to the tail.

3 Refinement workflow

Refinement is staged in five steps (Table 1) to decouple detector/sample geometry from physical scattering parameters, reducing correlations and improving identifiability. Table 1 summarizes the dominant image signatures that constrain each parameter group, while the

steps below describe the acquisition and fitting procedure. Additional workflow detail is provided in the supporting information (Sec. S1).

Table 1: Staged refinement workflow. For each step we list the primary parameter group and the dominant image signatures used to constrain it from area-detector images.

Step	Stage	Parameters	Dominant image signatures
1	Beam characterization (direct beam)	Beam widths $w_{x,z}$; divergences $\Delta\theta_{x,z}$	Spot centroid/width versus detector distance (instrument broadening).
2	Detector geometry calibration (3D powder standard)	Detector distance D; tilts (β, γ); in-plane rotation χ; beam center (x_0, y_0)	Ring center/ellipticity and orientation (pixel-to-$\mathbf{Q}$ mapping).
3	Sample alignment and goniometer misalignment (2D powder film)	Sample geometry $(\theta_i, \delta, z_B, z_S)$; goniometer misalignment (α, ψ); film (W, H, t)	Specular/off-specular positions; footprint truncation; absorption weighting.
4	Mosaicity from Bragg-peak shapes (two θ_i images)	Mosaic kernel (η, Γ, σ)	Arc/cap widths and specular off-condition tails.
5	Refinement of structure (Bragg + diffuse)	Scale; structure-factor parameters (o_a, $\mathbf{r}_a$, $\Delta\mathbf{r}_a$, $\mathbf{U}$); disorder (if present)	Relative Bragg intensities; diffuse along specular (Bragg rods) and between peaks along Q_z.

1. **Beam characterization (direct beam).** Acquire direct-beam images at several detector distances. Fit the spot centroid and width versus distance to determine the incident-beam phase-space parameters used by the diffraction model (Sec. 2).

2. **Detector geometry calibration (3D powder standard).** Measure a polycrystalline calibrant (e.g., hBN). Fit multiple Debye–Scherrer rings in detector coordinates and refine detector distance, tilts, in-plane rotation, and beam center so the rings forward-project to concentric circles.

3. **Sample alignment and goniometer misalignment (2D powder film).** Use a 2D powder film to seed alignment. With detector parameters fixed, refine sample alignment and goniometer misalignment by minimizing positional residuals between measured and simulated features spanning specular and off-specular regions. Include finite-footprint and absorption weighting using the film dimensions and thickness (Sec. 2).

4. **Mosaicity from Bragg-peak shapes.** Acquire two images at different θ_i and fit them simultaneously using shared mosaic parameters in the misorientation model of Sec. 2.3. Report Lorentzian FWHM $= 2\Gamma$, Gaussian FWHM $= 2\sqrt{2\ln 2}\,\sigma$, and mixing factor η.

5. **Refinement of structure (Bragg + diffuse).** With geometry and mosaicity fixed, refine an overall scale and structural parameters (occupancies, anisotropic Debye–Waller terms, and allowed static displacements) in the structure-factor intensity model (Eq. (1)) against the full 2D images (including specular). When disorder is present, include disorder parameters and constrain them primarily using the diffuse intensity along specular (Bragg rods) and between Bragg peaks along Q_z.

4 Results: ordered films

In this section we show representative fits for ordered MBE grown Bi_2Te_3 and Bi_2Se_3 films: 2D experimental images, 2D calculated images, and line cuts along Q_z with fits overlaid.

- 2D image data.
- 2D image calculation.
- Line cuts along Q_z with fitted curves.

5 Results: diffuse scattering in PbI_2

For disordered CVD grown PbI_2 we return to the Bragg sphere (cylinder) picture and introduce the Hendricks–Teller (H–T) stacking fault model used to describe the diffuse scattering. Polytype background and supporting derivations for the stacking-fault model are provided in the supporting information (Fig. S1; Secs. S2–S3).

The corresponding reciprocal-space geometry is sketched in Fig. 7.

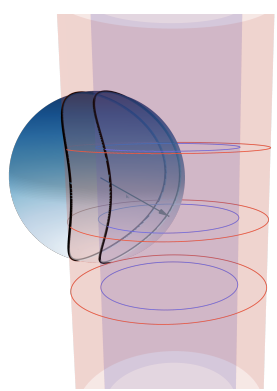

Figure 7: Reciprocal cylinders

- Motivate the model with experimental data.
- Outline the H–T model for PbI_2.
- Show resulting 2D images and line cuts with fits.

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